Physical Cumulative Exam
January 16, 1999

1. What are a) spherical rotors, b) symmetric rotors, and c) asymmetric rotors? Give an example for each case. (10 points)

2. State which of the following molecules may show a pure rotational microwave absorption spectrum: N_2, CO_2, OCS, H_2O, $CH_2{=}CH_2$, C_6H_6, CH_4, SF_6. (10 points; all or nothing)

3. State which of the following molecules may show a pure rotational Raman spectrum? N_2, CO_2, OCS, H_2O, $CH_2{=}CH_2$, C_6H_6, CH_4, SF_6. (10 points; all or nothing)

4. What are the degeneracies of the rotational energy levels of a) symmetric rotors, b) linear rotors, and c) spherical rotors? (10 points)

5. The rotational Raman spectrum of $^{35}Cl_2$ consists of a series of Stokes lines separated by $0.9752 cm^{-1}$. There is a similar series of anti-Stokes lines. What is the bond length of Cl_2? For the mass of Cl use 35 amu. (30 points)

6. For monomer CuBr at 1000 K in the vapor phase the following rotational transitions were found. For J=13 to J=14 at 84421.34 MHz; for J=14 to J=15 at 90449.25 MHz; for J=15 to J=16 at 96476.72 MHz. Calculate the bond length of CuBr. The masses are (in amu) Cu=63.6; Br=79.9. (30 points)

Help: for rotational energy levels $E = BJ(J+1)$, where the rotational constant is $B = \hbar / 4\pi cI$.
$\hbar$ = 1.054 E-34 Js; c = 2.998 E+8 m/s; 1.66 E-27 kg/amu.